# Sanaul Haque

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# IARCO RESEARCH PROPOSAL

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**Class/Grade/Year:** Class 8 2026

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**Submission Date:** 9/16/2026

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**Research Topic:** Flood-Conditioned, Asset-Augmented Shelter Allocation in Old Dhaka: A Cost-Minimizing Optimization Approach

Title: Flood-Conditioned, Asset-Augmented Shelter Allocation in Old Dhaka: A Cost-Minimizing Optimization Approach

**Research Problem**

4 Bangladesh is one of the most flood-exposed countries in the world, and seasonal waterlogging compounds this within Dhaka itself [8]. Old Dhaka is a particular case: some of the highest residential densities in South Asia, lanes often under three meters wide, and an irregular street pattern [5]. Straight-line accessibility measures overstate how many residents can reach a shelter, since they ignore the detours and dead ends that define real pedestrian movement; one Dhaka health-access study found a three-fold overestimate of emergency-care coverage once realistic travel conditions were modeled [4].

A deeper problem persists even in network-aware studies: every shelter-siting study reviewed for this proposal computes accessibility on the *dry-weather* street network, though a flood shelter is needed precisely when the network is not dry. Flood-disrupted road studies elsewhere show travel speed degrading non-linearly with water depth, with routes becoming effectively closed beyond a depth threshold [10], and pedestrian-instability research shows people lose the ability to walk safely at shallower depths than vehicles do [11]. A shelter that looks reachable on an ordinary day can become unreachable exactly when needed, a gap no cost-minimizing siting study addresses.

A second problem concerns which buildings count as candidate shelters. Bangladeshi and international practice alike restrict candidate sites to schools, community centers, and other government buildings [6]. Dhaka is popularly known as the “City of Mosques,” and Bangladesh’s Islamic Foundation has recorded over 250,000 mosques nationwide [12], a dense, evenly distributed building stock entirely outside the shelter-siting literature’s candidate pool. Mosques are open and staffed daily, unlike schools closed overnight and during vacations, are often multi-story, and require no land acquisition. A conceptual literature argues mosques suit emergency shelters [12], but no facility-location optimization has included them. Planners therefore lack a tool that prices flood-realistic accessibility while considering the low-cost sites already embedded in the community.

**Existing Literature**

The facility-location literature on disaster shelters is well developed. Sherali, Carter, and Hobeika [1] formulated the foundational location-allocation model for hurricane/flood evacuation. Gama, Santos, and Scaparra [2] extended this to a multi-period MILP, and Boromrit and Jamrus [3] incorporated priority scores and cost structure for Thai flood shelters. None is set in a South Asian megacity, and none draws its candidate pool beyond schools and government facilities. Dhaka-specific GIS studies on emergency healthcare [4] and public open space [5] show network-distance analysis is locally

feasible, but neither optimizes cost or site selection. A Saghata Upazila study [7] combines AHP mapping with routing to existing shelters, but is rural, treats sites as fixed, and does not minimize cost.

A separate literature computes road-network accessibility *under*, not before, flood conditions: Gangwal et al. degrade travel speed with flood depth for hospital access in Delaware [10], and Arrighi, Oumeraci, and Castelli establish a pedestrian, not vehicle, instability threshold [11]. Neither has been combined with a cost-minimizing siting MILP or applied in South Asia. A conceptual literature proposes mosques as emergency shelters [12], but never as a candidate-site input to a formal model. A third literature documents Bangladesh's Cyclone Preparedness Programme (CPP), a 76,000-volunteer, UN-cited institution that already warns via mosque loudspeakers and documents informal sheltering in nearby strong buildings [15], [16], but has not translated that network into a siting objective. The gap this proposal addresses combines four things never combined before: a cost-minimizing MILP [1]–[3], a flood-conditioned accessibility constraint [10], [11], an expanded candidate pool [12], and CPP's warning infrastructure [15], [16], applied to a named Old Dhaka ward to decompose each element's cost contribution.

## Research Question

***Research question:*** In a defined high-density ward of Old Dhaka, (a) what configuration of flood shelters, drawn from an expanded candidate pool that includes mosques and madrasas alongside schools and community centers, minimizes total public shelter-provision cost, subject to the constraint that every resident can reach an open shelter within T minutes under flood-conditioned, rather than dry-weather, walking conditions; and (b) how much of the resulting cost saving, relative to a conventional schools/community-centers-only, dry-weather baseline, is attributable to the expanded candidate pool versus the more realistic accessibility constraint?

**Objectives:** (O1) assemble population, candidate-site, street-network, and flood-depth data for the chosen ward; (O2) formulate and solve a baseline MILP (conventional sites, dry-weather network) and an augmented MILP (expanded sites, flood-conditioned network); (O3) run sensitivity analysis across walk-time standards T and decompose the cost gap between models by source.

## Methodology

### Study Area and Unit of Analysis

The analysis will focus on Bangshal, a Dhaka South City Corporation ward in the Old Dhaka core that DSCC officials and repeated multi-year press investigations name as one of the city's most chronic waterlogging hotspots, with stagnant water reported to persist for 15–20 hours after moderate rainfall even after DSCC-claimed mitigation spending

[13], [14]. Bangshal also fits the density and street-geometry profile motivating this proposal, making it a concrete, verifiable rather than illustrative choice of ward. Demand will be aggregated to mahalla (neighborhood) centroids to keep the model computationally tractable while preserving intra-ward spatial variation.

### Data Sources

19

**Population/demand weights:** Bangladesh Bureau of Statistics, Population and Housing Census 2022, ward/mahalla-level tables [8].

**Baseline candidate sites:** government schools, community centers, and other public buildings identified from Dhaka South City Corporation facility records and BWDB flood-shelter registries.

**Expanded candidate sites and willingness screening:** mosques and madrasas from OpenStreetMap place-of-worship/religious-education tags, cross-validated against Islamic Foundation registries and DSCC records, screened for structural suitability [12], and assigned a site willingness score from a structured engagement survey with the local mosque committee (modeled on CPP's community-engagement approach), so consent is measured, not assumed [16].

**Warning-broadcast capability:** loudspeaker equipment at each mosque, field-verified for presence and range, building on mosque microphones' established role in CPP's warning network [15], [16].

**Street network:** OpenStreetMap data via OSMnx, manually corrected against satellite imagery for lanes known to be mis-mapped or informally narrowed, a documented issue in unplanned Dhaka [5].

**Flood-depth data:** DSCC waterlogging-complaint records plus a small participatory high-water-mark survey via CPP's volunteer model, chosen over satellite (SAR) imagery, which loses reliability at Old Dhaka's building density. These readings calibrate the depth-disruption function [10].

**Cost data:** fixed opening/retrofit costs estimated from BWDB Multipurpose Disaster Shelter Project unit-cost benchmarks [9], scaled by building type and capacity; mosque/madrasa retrofits (signage, WASH upgrades) should cost less than new construction, since the structure and staffing already exist. Where a site-specific figure is unavailable, this is flagged as a modeling assumption and tested, not hidden, in the sensitivity analysis below.

### Model Formulation

21

Let I be the set of candidate shelter sites, partitioned into $I^s$ (schools and community centers) and $I^M$ (mosques and madrasas), so $I = I^s \cup I^M$, and let J be the set of demand nodes. Let $f_i$ be the fixed cost of opening site $i \in I$, $c_{ij}$ the allocation cost of serving node

$j \in J$ from site i, $pop_j$ the population at node j, $cap_i$ the capacity of site i, $d_{ij}$ the shortest-path walking time from j to i on the street network, and T the maximum allowable accessibility standard, in minutes. Let $reach_{ij} \in \{0,1\}$ indicate that site i has field-verified warning-broadcast (loudspeaker) capability and node j lies within its audible range, and β the minimum required share of ward population with warning-broadcast coverage. Decision variables: $x_i \in \{0,1\}$, equal to 1 if site i is opened; $y_{ij} \in \{0,1\}$, equal to 1 if node j is assigned to site i; $w_j \in \{0,1\}$, equal to 1 if node j is reached by warning broadcast from an opened site.

$$\text{minimize} \quad \Sigma_{i \in I} f_i x_i + \Sigma_{i \in I} \Sigma_{j \in J} c_{ij} y_{ij} \tag{1}$$

$$\text{subject to:} \quad \Sigma_{i \in I} y_{ij} = 1, \quad \forall j \in J \tag{2}$$

$$y_{ij} \leq x_i, \quad \forall i \in I, j \in J \tag{3}$$

$$\Sigma_{j \in J} pop_j \, y_{ij} \leq cap_i \, x_i, \quad \forall i \in I \tag{4}$$

$$d_{ij} \, y_{ij} \leq T, \quad \forall i \in I, j \in J \tag{5}$$

$$\Sigma_{i \in I^s} x_i \geq \alpha \, \Sigma_{i \in I} x_i \tag{6}$$

$$w_j \leq \Sigma_{i \in I} reach_{ij} \, x_i, \quad \forall j \in J \tag{7}$$

$$\Sigma_{j \in J} pop_j \, w_j \geq \beta \, \Sigma_{j \in J} pop_j \tag{8}$$

$$x_i, \, y_{ij}, \, w_j \in \{0,1\} \tag{9}$$

Constraint (2) assigns each demand node to one opened shelter; (3) restricts assignment to opened sites; (4) enforces capacity; (6) sets an inclusivity floor: a share α of opened shelters must be secular ($I^s$), so residents uncomfortable at a religious site, including many women and religious minorities, always have a non-mosque option. Constraints (7)–(8) extend CPP's mosque-loudspeaker warning network into urban siting, valuing each mosque for two things at once: walking accessibility and loudspeaker warning reach. That dual value is what separates this from simply adding mosques to a school-only list.

Constraint (5) is evaluated two ways to isolate each idea's contribution. In baseline M0, I is restricted to $I^s$, $d_{ij}$ is dry-weather walking time, and constraints (7)–(8) drop out. In augmented M1, $I = I^s \cup I^M$, $d_{ij}$ is flood-conditioned walking time (segment speed degraded by a depth-disruption function, with segments beyond the pedestrian instability threshold removed), and the broadcast floor applies. Two intermediate variants complete a 2×2 factorial design that decomposes the M0–M1 cost gap by idea. Only mosques and madrasas clearing a minimum committee-willingness score enter $I^M$. The model runs in Python (PuLP/CBC, Gurobi where available), with shortest-path matrices pre-computed in OSMnx/NetworkX.

**TABLE I**

**MODEL NOTATION SUMMARY**

| Symbol | Definition |
|---|---|
| I, J | Sets of candidate shelter sites and demand (mahalla) nodes |
| $I^s$, $I^M$ | Subsets of I: schools/community centers; mosques/madrasas ($I = I^s \cup I^M$) |
| $f_i$ | Fixed cost of opening shelter i |
| $c_{ij}$ | Allocation/operating cost of serving node j from site i |
| $pop_j$ | Population at demand node j |
| $cap_i$ | Capacity of shelter i |
| $d_{ij}$ | Walking time from j to i: dry-weather (M0) or flood-conditioned (M1) |
| T | Accessibility standard: maximum allowable walk time (minutes) |
| α | Minimum required share of opened shelters that are secular ($I^s$) sites |
| $reach_{ij}$ | 1 if site i is broadcast-capable and node j is within its audible range |
| β | Minimum required share of population with warning-broadcast coverage |
| $x_i$, $y_{ij}$, $w_j$ | Binary decision variables (open site i; assign j to i; broadcast-cover j) |

## Sensitivity Analysis

All four model variants will be re-solved for $T \in \{10, 15, 20, 25, 30\}$ minutes (and additional values as needed) to trace minimum achievable cost as a function of the accessibility standard, producing the core deliverable of this proposal: comparative cost–accessibility trade-off curves for the baseline and augmented models (Fig. 1). The broadcast-coverage floor β will also be varied to trace how much of the augmented model's cost advantage survives once broadcast coverage is required. Where assumptions carry meaningful uncertainty, the model will be re-solved under low- and high-severity flood and cost scenarios to test the sensitivity of the curves themselves.

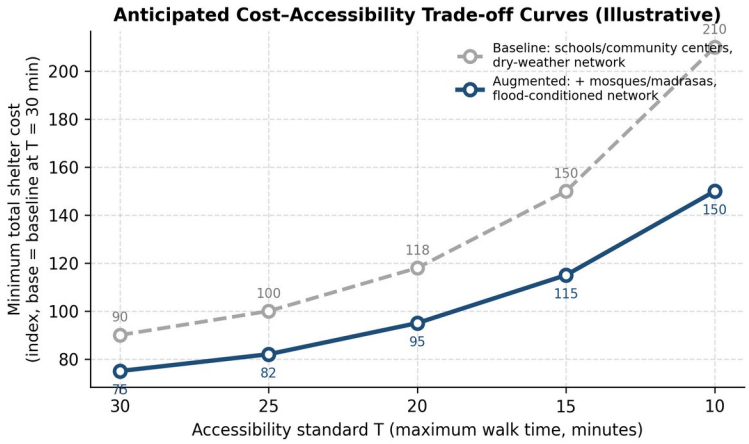



*Fig. 1. Anticipated cost–accessibility trade-off curves for baseline (M0) and augmented (M1) models. Values are illustrative; actual curves will come from solving Eqs. (1)–(9) once ward-level data are assembled.*

### Expected Outputs

(i) baseline and augmented shelter configuration maps for a common T; (ii) comparative trade-off curves across T for all four model variants (Fig. 1); (iii) a decomposition of cost savings by source; (iv) a sensitivity table for the accessibility standard, broadcast target, and flood severity.

### Research Topic

Optimization-based shelter siting is an established technique [1]–[3], but this proposal combines three overlooked ideas for the first time: flood-conditioned accessibility [10], [11], an expanded mosque-based candidate pool [12], and CPP's loudspeaker network as an explicit siting objective [15], [16]. None is technically exotic; that is the point. Flood frequency in Bangladesh is projected to rise with climate change, and Old Dhaka's street network is aging and largely fixed, so a siting decision made today is locked in for decades, making it worth accounting for that geometry, and the institutions embedded in it, now. Separating how much cost comes from each of these three factors gives DSCC and BWDB a decomposition directly transferable to other flood-prone, mosque-dense cities across South Asia.

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